Technoverse Hackathon

2026

Team Name: 2Bits
Domain: Retail
Theme: Supply Chain Resilience

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XYRA— Predict the rush. Never run out | Business Plan (1/2)

WHY

Explain the Problem

Problem Definition/Business Scenario: India's retail industry loses around ₹1.2T each year because of two factors, **stockouts (empty shelves that deter customers)** and **dead stock (oversupply of items that occupy cash flow).** *The root cause is not poor management — it is zero visibility into what is coming, Demand spikes from Local Events. 68% of small and medium-sized retailers in India continue to practice manual stock counting and intuition.*
Scope of Problem: Concentrated on small and medium-sized retail chains (with 1-20 branches) in India that operate in grocery, pharmacy, and FMCG sectors. The application covers three main areas — **demand forecasting, identification of disruptions in supply chain, and reordering decisions.** *It will not substitute ERP systems since it can operate both with or without them.*
Target Users/Customers: Store Managers (primary users), Chain Owners, Procurement teams (reordering actions)

How

Explain the Solve

Solution Overview: **XYRA** is an AI-powered supply chain assistant that **ingests a store's historical sales CSV**, **pulls live local event data** by pincode, **monitors weather disruptions**, and **uses an reasoning engine to predict demand spikes per product category**. It then identifies the best available vendor, notifies them, and places a reorder request — all pending one-click manager approval.
No complex setup, no IT team required.
Innovation: Unlike generic inventory tools adapted from enterprise software, **XYRA** is the only solution that connects local events + real-time weather + sales patterns + supplier status to trigger context-aware reorder decisions with one-click manager approval.
Purpose-built for Indian retail.
Market Potential:

- 15M+ small & mid retailers in India
- ₹1.2T lost annually to supply chain inefficiencies
- AI in Retail Inventory Management market: $6.7B in 2024 → $33.6B by 2034 (17.7% CAGR)
- Scalable across grocery, pharmacy, FMCG, and quick-commerce stores

WHAT

Value proposition

Primary Benefits: Automatically predicts demand spikes, selects the best available vendor by product type, and places the reorder with a single manager approval — eliminating manual effort entirely.
Time and Cost Saving: Cuts reordering time from hours to minutes. Reduces stockouts by up to 35% and excess inventory by up to 28%.
Efficiency and Flexibility: No more manual stock checks or vendor phone calls. **XYRA** instantly recommends the right action for any product category, adaptable to any store size or branch layout.

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Financials & Timelines | Business Plan (2/2)

Investments

What does it take & How much does it cost to solve?

Cloud Infrastructure: ~₹8,000–12,000/month [Covers hosting of backend APIs and frontend dashboard on AWS, ensuring scalability and reliability**]**
Events & Weather APIs: Free tiers are sufficient for MVP; paid plans (~₹2,000/month) required as the usage scales by multiple stores.
Development Effort: MVP can be built in 2 weeks; production-ready system achievable within 6 weeks.
Total MVP Cost: Under ₹15,000 **[**Covers end-to-end development and initial deployment of a fully functional prototype.**]**
Deployment & Integration: No legacy system dependency — operates standalone via CSV uploads or lightweight POS integration, enabling quick adoption.
Maintenance & Scaling: Covers system monitoring, updates, bug fixes, and scaling support as store count grows

₹1.2T/year lost globally (stockouts + overstock) → XYRA recovers ₹400–₹500B/year (35–40%)
35% fewer stockouts → ₹350B+ revenue regained globally, 20% fewer unnecessary orders
Emergency buying → ₹50B+ saved annually,15–25% inventory holding cost savings
Vendor dependency ↓ → faster procurement, lower delays, reduced churn from stockouts
IF NOT SOLVED → **₹1.2T/year continues to leak globally ,**
Peak-demand stockouts → 30–40% daily revenue loss ,
Poor forecasting → continuous overstock + stockout cycle

35-40%
[stockouts + overstock]
XYRA recovers”

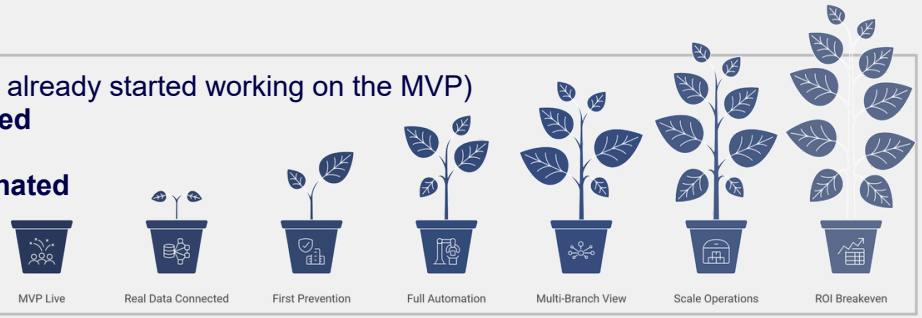
Returns

Quantify the benefits & What if I don’t solve?

Timelines

Time to realize benefits

Week 2 → **MVP live → first demand predictions ~** (We have already started working on the MVP)
Week 3–4 → **Real data connected → stockout risks identified**
Month 1–2 → **First prevention → ₹25K–40K saved/event**
Month 2 → **Full automation → 2 hrs/day manual work eliminated**
Month 3 → **Multi-branch view → ~20% fewer new orders**
Month 6 → **Scale (50 stores) → ₹1.5Cr+ annual recovery**
Month 2–3 → **ROI breakeven (8–10 weeks)**



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Why This Problem Is Worth Solving

Market Data

1. **$6.70B** → **$33.60B** Global AI in Retail Inventory Market projected to grow 5x in 10 years at **17.7% CAGR** — proving this is not a trend, it is the future of retail. (InsightAce Analytic, 2024)
2. **15M+** Small and mid-size retailers in India represent the largest underserved retail segment in the world — yet have access to **zero affordable AI tools**.
3. **₹1.2 Trillion** Lost every single year in Indian retail supply chain inefficiencies — **stockouts**, **overstocking**, and missed demand opportunities.

Cost of Inaction

1. Missing a single festival demand spike costs a retailer **30–40%** of that entire day's revenue — **permanently unrecoverable.**
2. Walmart's event and weather-aware forecasting reduced stockouts by **30%** — **XYRA** brings this exact capability to small Indian retailers

Research Backing

[1] O. R. Amosu, P. Kumar, Y. M. Ogunsuji, S. Oni, and O. Faworaja, **"AI-driven demand forecasting: Enhancing inventory management and customer satisfaction,"** World Journal of Advanced Research and Reviews, vol. 23, no. 2, pp. 708–719, Aug. 2024,

[2] S. K. R. Malikireddy, **"Enhancing retail supply chain resilience with generative AI,"** World Journal of Advanced Engineering Technology and Sciences, vol. 9, no. 1, pp. 399–409, Jun. 2023,

[3] P. Yadav, **"Demand forecasting in retail using machine learning and big data,"** International Journal of Advanced Research in Computer and Communication Engineering, vol. 14, no. 2, pp. 274–278, Feb. 2025,

[4] J. Jones, **"AI-driven demand forecasting in supply chains: A qualitative analysis of adoption and impact,"** Preprints.org, Jan. 2025,

[5] A. Chauhan, **"Retail store inventory forecasting dataset," Kaggle,2024.**
[Online Available: https://www.kaggle.com/datasets/anirudhchauhan/retail-store-inventory-forecasting-dataset]

System Architecture ***Live Link*** ***Github***

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Thank you

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